

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/Toyota.csv")
print(df)
```

```

      Unnamed: 0  Price  Age   KM FuelType  HP  MetColor  Automatic   CC  \
0              0  13500  23.0  46986   Diesel  90      1.0         0  2000
1              1  13750  23.0  72937   Diesel  90      1.0         0  2000
2              2  13950  24.0  41711   Diesel  90      NaN         0  2000
3              3  14950  26.0  48000   Diesel  90      0.0         0  2000
4              4  13750  30.0  38500   Diesel  90      0.0         0  2000
...           ...    ...    ...    ...    ...    ...    ...    ...    ...
1431          1431   7500   NaN  20544   Petrol  86      1.0         0  1300
1432          1432  10845   72.0    ??   Petrol  86      0.0         0  1300
1433          1433   8500   NaN  17016   Petrol  86      0.0         0  1300
1434          1434   7250   70.0    ??      NaN  86      1.0         0  1300
1435          1435   6950   76.0     1   Petrol 110      0.0         0  1600
```

```

      Doors  Weight
0       three   1165
1         3   1165
2         3   1165
3         3   1165
4         3   1170
...      ...    ...
1431         3   1025
1432         3   1015
1433         3   1015
1434         3   1015
1435         5   1114
```

[1436 rows x 11 columns]

```
df.head(5)
```

	Unnamed: 0	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors	Weight
0	0	13500	23.0	46986	Diesel	90	1.0	0	2000	three	1165
1	1	13750	23.0	72937	Diesel	90	1.0	0	2000	3	1165
2	2	13950	24.0	41711	Diesel	90	NaN	0	2000	3	1165
3	3	14950	26.0	48000	Diesel	90	0.0	0	2000	3	1165
4	4	13750	30.0	38500	Diesel	90	0.0	0	2000	3	1170

```
df.isnull().sum()
```

```

      0
Unnamed: 0    0
Price         0
Age         100
KM           0
FuelType     100
HP           0
MetColor     150
Automatic     0
CC           0
Doors         0
Weight        0
```

dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1436 entries, 0 to 1435
```

```
Data columns (total 11 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Unnamed: 0    1436 non-null     int64
1   Price         1436 non-null     int64
2   Age           1336 non-null     float64
3   KM            1436 non-null     object
4   FuelType      1336 non-null     object
5   HP            1436 non-null     object
6   MetColor      1286 non-null     float64
7   Automatic     1436 non-null     int64
8   CC            1436 non-null     int64
9   Doors         1436 non-null     object
10  Weight        1436 non-null     int64
dtypes: float64(2), int64(5), object(4)
memory usage: 123.5+ KB
```

df.describe()

	Unnamed: 0	Price	Age	MetColor	Automatic	CC	Weight
count	1436.000000	1436.000000	1336.000000	1286.000000	1436.000000	1436.000000	1436.000000
mean	717.500000	10730.824513	55.672156	0.674961	0.055710	1566.827994	1072.45961
std	414.681806	3626.964585	18.589804	0.468572	0.229441	187.182436	52.64112
min	0.000000	4350.000000	1.000000	0.000000	0.000000	1300.000000	1000.00000
25%	358.750000	8450.000000	43.000000	0.000000	0.000000	1400.000000	1040.00000
50%	717.500000	9900.000000	60.000000	1.000000	0.000000	1600.000000	1070.00000
75%	1076.250000	11950.000000	70.000000	1.000000	0.000000	1600.000000	1085.00000
max	1435.000000	32500.000000	80.000000	1.000000	1.000000	2000.000000	1615.00000

df.dtypes

```

      0
Unnamed: 0    int64
Price         int64
Age           float64
KM            object
FuelType      object
HP            object
MetColor      float64
Automatic     int64
CC            int64
Doors         object
Weight        int64

dtype: object
```

df.dropna()

	Unnamed: 0	Price	Age	KM	FuelType	HP	MetColor	Automatic	CC	Doors	Weight
0	0	13500	23.0	46986	Diesel	90	1.0	0	2000	three	1165
1	1	13750	23.0	72937	Diesel	90	1.0	0	2000	3	1165
3	3	14950	26.0	48000	Diesel	90	0.0	0	2000	3	1165
4	4	13750	30.0	38500	Diesel	90	0.0	0	2000	3	1170
5	5	12950	32.0	61000	Diesel	90	0.0	0	2000	3	1170
...	...	...	...	...	...	...	...	...	...	...	...
1425	1425	7950	80.0	??	Petrol	86	1.0	0	1300	4	1000
1429	1429	8950	78.0	24000	Petrol	86	1.0	1	1300	5	1065
1430	1430	8450	80.0	23000	Petrol	86	0.0	0	1300	3	1015
1432	1432	10845	72.0	??	Petrol	86	0.0	0	1300	3	1015
1435	1435	6950	76.0	1	Petrol	110	0.0	0	1600	5	1114

1111 rows × 11 columns

```

mean_val=df['Age'].mean()
df['Age']=df['Age'].fillna(mean_val)
mean_val=df['MetColor'].mean()
df['MetColor']=df['MetColor'].fillna(mean_val)
df.isnull().sum()

```

```

      0
Unnamed: 0    0
Price         0
Age          0
KM           0
FuelType    100
HP           0
MetColor     0
Automatic    0
CC           0
Doors        0
Weight       0

```

dtype: int64

```

df['Price']=df['Price'].astype('float')
df.dtypes

```

0	
Unnamed: 0	int64
Price	float64
Age	float64
KM	object
FuelType	object
HP	object
MetColor	float64
Automatic	int64
CC	int64
Doors	object
Weight	int64